Assignment - 2

Multiple Linear Regression

$$\theta = (X^T X)^{-1} X^T Y$$

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 135 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix} \qquad Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$XX^T = \begin{bmatrix} 5 & 45 & 66 & \cancel{960}\ 960 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 960 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$\cancel{x^T}\ X^T Y = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 116235 \\ \cancel{17435} \end{bmatrix}$$

$|X^T X| \neq 0$

$\cancel{X^T X}$

$$(X^T X)^{-1} \approx \begin{bmatrix} 76.9017 & 18.2148 & -3.0043 & -1.104 \\ 12.2148 & 4.5863 & -0.7241 & -0.2744 \\ -3.6043 & -0.7241 & 0.1602 & 0.0467 \\ -1.1043 & -0.2744 & -0.0467 & 0.0167 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{bmatrix} -99.911 \\ 1.217 \\ 2.417 \\ 0.906 \end{bmatrix}$$

Final Equation

$$y = -99.911 + 1.217x_1 + 2.417x_2 + 0.906x_3$$

$x_1, x_2, x_3 \rightarrow$ independent variab

$\cancel{x_3 \rightarrow}$ $y \rightarrow$ Dependent

$x_1 \Rightarrow$ windspeed

$x_2 \Rightarrow$ blade angle $\quad y \rightarrow$ Power output

$\rightarrow$ Rotor speed

given

$$y = -99.911 + 1.217(9) + 2.417(11) + 0.906(100)$$

$$= -99.911 + 10.953 + 26.587 + 90.6 \Rightarrow y = 28.23$$

$$\boxed{y = 28.2290 \approx \underline{\underline{28.23}}}$$